Problem 1.10.36

Among those people who are infected with a certain virus, 32% have strain A, 59% have strain B, and the remaining 9% have strain C. Furthermore, 21% of people infected with strain A of the virus exhibit symptoms, 16% of people infected with strain B of the virus exhibit symptoms, and 63% of people infected with strain C of the virus exhibit symptoms.

(a) If a person has the virus and exhibits symptoms of it, what is the probability that they have strain C?

(b) If a person has the virus but doesn't exhibit any symptoms of it, what is the probability that they have strain A?

(c) What is the probability that a person who has the virus does not exhibit any symptoms of it?

Problem 1.10.37

The marketing division of a company profiles its potential customers and grades them as either likely or unlikely purchasers. Overall, 16% of the potential customers are graded as likely purchasers. In reality, 81% of the potential customers graded as likely purchasers actually make a purchase, while only 9% of the potential customers graded as unlikely purchasers actually make a purchase. If somebody made a purchase, what is the probability that they had been graded as a likely purchaser?

Problem 2.1.6

A fair coin is flipped three times. On the first flip, a player gains \$1 for heads and loses \$1 for tails. On the second flip, the gain or loss is \$2 for heads or tails, respectively. On the third flip, the payout is \$3 for heads and -\$3 for tails. Let X represent the total net payout across all three flips.

(a) Determine the probability mass function of X.

(b) Derive the cumulative distribution function of X.

(c) Find the most probable value of X.

Problem 2.1.7

Suppose a television set receives a quality score computed as the product of its picture rating and appearance rating. The picture ratings are assigned values of 4 (perfect), 2 (good), 1 (satisfactory), and 0 (failed). The appearance ratings are assigned values of 3 (perfect), 2 (good), 1 (satisfactory), and 0 (failed). For instance, a television with a perfect picture and good appearance achieves a quality score of $4 \times 2 = 8$.

(a) Formulate the probability mass function of the quality score.

(b) Formulate the cumulative distribution function of the quality score.

(c) Identify the mode (most likely value) of the quality score.

Additionally, suppose an item is rejected (not shipped) if either attribute receives a score of 0, or if both attributes receive a score of 1. Explain how the probability of non-shipment can be determined directly using the cumulative distribution function.

Problem 2.1.8

Four cards marked with values \$1, \$2, \$3, and \$6 are placed in a box. A player pays \$4 to draw two cards at random without replacement and receives an amount equal to the sum of the values on the drawn cards. Find the probability mass function and the cumulative distribution function for the net profit (total payout minus the \$4 entry fee).

Problem 2.1.9

Out of five warehouses managed by a firm, exactly two carry a specific item in stock. A representative contacts the warehouses sequentially in a random order until locating one with available stock. Let X denote the number of inquiries placed until stock is found. Determine the probability mass function and the cumulative distribution function for X.

Problem 2.1.10

Consider a discrete random variable X taking values in the set of positive integers $\{1, 2, 3, \dots\}$.

(a) Can $P(X = i) = \frac{c}{2^i}$ define a valid probability distribution for an appropriate constant c?

(b) Can $P(X = i) = \frac{c}{i}$ define a valid probability distribution for an appropriate constant c?

Problem 2.1.11

An advisor has six available time slots for scheduling client appointments: three on Monday and three on Tuesday. Incoming clients select among all currently unreserved slots with equal probability. Let X represent the total number of bookings completed across both days at the exact moment Monday's calendar becomes fully booked.

(a) Identify the state space (possible values) of the random variable X.

(b) Compute the probability mass function and cumulative distribution function of X.